

## **PSAT coding sheet 16/3/26**

- 1) Videos and photos should be recorded during off peak to prevent obscuring views.
- 2) It is still useful to code as 10m interval as there could be observations like potholes, obstacles, unless it is along rural roads.  
Any bigger interval might miss important features like driveway.
- 3) If there are more than 1 coder, coders have to discuss on their coding regularly to iron out discrepancies.
- 4) For each photo, code the most prominent type (path or junction) or look at the photos in front or back to see if junction has already been coded; if no, then code this photo as junction.

| Attribute                                                                    | Code                  | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Area Type</b><br>Urban<br>Suburban<br>Rural<br>Industrial<br>Recreational | 1<br>2<br>3<br>4<br>5 | <p>Select the predominant area type to assist with analysis. Standard options may be altered to better meet the local context. Do not change area type over short lengths (less than 100m)</p> <p><b>Urban</b></p> <ul style="list-style-type: none"> <li>• High-density residential or commercial development</li> <li>• High street connectivity, frequent intersections</li> <li>• Typically within city centres or dense neighbourhoods</li> <li>• OSM <code>landuse=residential</code>, <code>commercial</code>, <code>retail</code></li> </ul> <p><b>Suburban</b></p> <ul style="list-style-type: none"> <li>• Lower density residential areas with moderate connectivity</li> <li>• Typically outer city ring areas or commuter towns</li> <li>• Detached housing, local shops, fewer intersections</li> </ul> | <p>Most of our cycling paths should be classified as outer urban (residential), except those in central area (inner urban)</p> <p><b>Default: Suburban</b></p> <p>Geospace: LTA Parking standard Zone 1 comprises of city (restricted zone &amp; Marina Bay) → Urban</p> <p>Geospace: URA Amendment to Master Plan 2019 Land Use layer</p> <ul style="list-style-type: none"> <li>- Agriculture, Cemetery, Open space, Water Body, Special use → Rural</li> <li>- Business 1, Business 2, Transport Facilities → Industrial</li> <li>- Beach area, park, sports &amp; recreation → recreational</li> </ul> |

| Attribute | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |      | <p><b>Rural</b></p> <ul style="list-style-type: none"> <li>Predominantly agricultural or undeveloped land</li> <li>Sparse or linear development</li> <li>Few intersections, long distances between buildings</li> <li>OSM <code>landuse=farmland, forest, grass</code></li> </ul> <p><b>Industrial</b></p> <ul style="list-style-type: none"> <li>Warehouses, factories, logistics centres</li> <li>Zoning or land use as <code>landuse=industrial</code></li> <li>Often co-located with heavy vehicle routes</li> </ul> <p><b>Recreational</b></p> <ul style="list-style-type: none"> <li>Parks, trails, waterfronts, sporting facilities</li> <li><code>landuse=recreation_ground, leisure=park, natural=*</code></li> </ul> <p>Common for off-road paths, shared-use facilities away from roads</p> |  <p>Suburban takes precedence if there is an overlap</p> <p><i>To include an offset of 10m (to improve accuracy); CPN is usually about 1.5m away from the boundary line of land use layer; at development accesses, might not be able to capture, hence to do manually.</i></p>  <p>[GIS]</p> |

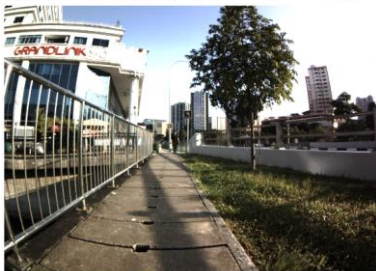
| Attribute                        | Code | Coding guidance                                                                                                                                                | Local guidance                                                                                                           |
|----------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Facility type</b><br>Sidewalk | 1    | An off-street path intended for pedestrian use only                                                                                                            | Footpath                              |
| Multi-use path                   | 2    | An off-street path which is intended for use by bicycles/light vehicles and pedestrians. Multiuse paths are generally wider with signage indicating it as such | Shared 2.5m path                      |
| Off-road bicycle path            | 3    | An off-street path which is intended for use by bicycles/light vehicles only                                                                                   | Most of our off-road cycling paths  |

| Attribute               | Code | Coding guidance                                                                                                                                  | Local guidance                                                                                                                                                                                                                                                                                       |
|-------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| On-road bicycle lane    | 4    | An on-street lane intended for use by bicycles/light vehicles, which is out of the main path of large, motorized vehicles and marked accordingly |  <p>Two solid red lines used in city area<br/>Eg: TMCR bicycle lane, demarcated bicycle crossing</p>                           |
| Road shoulder           | 5    | Space on the street (but out of the direct path of large, motorized vehicles) used by bicyclists/light vehicles. May be unmarked                 | <p>Eg: West Camp Road cycling lane, cyclists sharing bus lane; development access, zebra crossing, signalized crossing</p>   |
| Mixed traffic road lane | 6    | Bicycles/light vehicles use the same street space/path as large, motorized vehicles (only this category has a risk factor)                       |  <p>Default: Sidewalk<br/>[CV]</p>                                                                                                                                                                              |

| Attribute                          | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Facility access</b><br>Adequate | 1    | The facility is accessible and fit for purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Bicycle wheeling ramp on POB is coded as adequate, unless you observe that there are cyclists crossing the road and not using the wheeling ramp. Chicanes is coded as adequate as cyclists still can go through (code fixed obstacles present). Construction water filled barricade affect clear width but still passable, code as adequate</p>  |
| Inadequate                         | 2    | <p>Due to congestion, obstructions or other issues, some or all bicyclists and light vehicle users are forced to use alternative route (adjacent paths or roadway). E.g. very high ped flows, very poor surface quality, bollards/fencing which reduce path width for cargo bicycles, lack of ramps.</p> <p><i>Inadequate affects VB risk.</i></p> <p>Flag “inadequate” if congestion or blockages detected from:</p> <ul style="list-style-type: none"> <li>- Obstruction reports</li> <li>- Counterflow or avoidance behaviour</li> </ul> <p>Vehicle/bike/pedestrian volume overload</p> | <p>Hoarding by construction or path too crowded (till it forces cyclists to go onto the road) is coded as inadequate; no bypass path behind bus stop forces cyclist to go in front of bus stop or on the road</p>                                                                                                                                 |



| Attribute                                          | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                 | Local guidance                                                                                                                                                                                                       |
|----------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Loose or slippery surface</i></b><br>Present | 1    | Record if there is loose gravel, metal plates, wet leaves, a crossfall or any other characteristic present which could result in a bicyclist or light vehicle to lose traction<br><br>Surface = gravel, leaves, metal plates, etc. occupying more than 1m <sup>2</sup> of a 10m coding segment (this may be several smaller patches which amount to more than 1m <sup>2</sup> ) | There could be some stretches with soil erosion from the adjacent planter box/area<br>Fallen leaves if a lot coded as present<br> |
| Not present                                        | 2    |                                                                                                                                                                                                                                                                                                                                                                                 | [CV still not ready yet]<br>To wait for grip tester results from PATH                                                                                                                                                |
| <b><i>Tram or train rails</i></b><br>Present       | 1    | Record if tram rails are present on or in the vicinity of the facility which may be crossed by a bicyclist or light vehicle                                                                                                                                                                                                                                                     | There is no such path/road in Singapore → 2 (Default)                                                                                                                                                                |
| Not present                                        | 2    |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                      |

| Attribute                                                    | Code | Coding guidance                                                                                                                                                                                                                                  | Local guidance                                                                                                                                                                                                                                          |
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| <b>Major surface deformation or drain opening</b><br>Present | 1    | Record if there is any major surface deformation which could result in a bicyclist or light vehicle to lose balance. Examples include potholes, tree roots, major cracking, drain openings or tarmac edges or ramps with a vertical edge of >3cm |   <p>Uneven drain covers – depends if it will cause a severe or fatal crash.</p> |
| Not present                                                  | 2    |                                                                                                                                                                                                                                                  | [Pending PATH's defects layer]                                                                                                                                                                                                                          |



| Attribute                                    | Code | Coding guidance                                                                                                                                                                                                  | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Fixed obstacle on facility</b><br>Present | 1    | Record if there is one or more fixed obstacles on the facility that is not movable. Examples include poles, fences, bollards, street furniture, electrical boxes, bus stops, signages, bike racks, shelters, etc | Record if covered linkway poles/OG boxes/ lamp post/VIG posts are within the path width, chicanes. Outgrown vegetation that encroaches onto path. <div>       </div> |
| Not present                                  | 2    |                                                                                                                                                                                                                  | [CV]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Attribute                                        | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Local guidance                                                                                                                                                                            |
|--------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-fixed obstacle on facility</b><br>Present | 1    | Record if there is one or more non-fixed obstacle/s on the facility. Examples include rubbish bins, parked cars or bicycles, roadwork signs or device, or other temporary items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Record if there is any non-fixed devices, rubbish bins, motorcycles, advertisement panels, etc<br><br> |
| Not present                                      | 2    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [CV]                                                                                                                                                                                      |
| <b>Line of sight</b><br>Adequate                 | 1    | Record if there are any visual obstructions on or around the path which would prevent a cyclist from seeing (or being seen by) an approaching vehicle on a driveway or side road, or another cyclist or pedestrian. Visual obstructions may include shrubs (such as hedges), fences, walls etc.<br><br>Check visibility from eye height (1.2-1.5m) along the path and toward any driveway, side road, bend or crossing point for <ul style="list-style-type: none"> <li>• Objects blocking the view withing approximately 10-20m of the conflict point</li> <li>• Continuous objects above 0,6m-0.8m height (e.g. hedges, fences, walls, parked vehicles, dense vegetation) within 1m of the path edge</li> </ul> |                                                                                                       |
| Inadequate                                       | 2    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                      |

| Attribute                                                                            | Code                                           | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                            | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Delineation</b><br>Present<br><br><br><br><br><br><br><br><br><br><br>Not present | 1<br><br><br><br><br><br><br><br><br><br><br>2 | Record if clear edge and central marking (for two-way facilities) are present on the facility.<br><br><br><br><br><br><br><br><br><br><br>Record as 'not present' if they are absent, only partially present or are in poor condition. Delineation for on-road facilities should only be recorded if provided for a bike lane or road shoulder. Record as 'not present' for mixed traffic road lanes (even if delineation is provided for general traffic) | <p>For most of our newly paved cycling/ shared path, for zebra crossings, signalized crossing</p> <p>Record if cycling/ shared path markings have faded (both sides faded that it cannot be recognized as shared path). Record if it is development access (no delineation). For West camp road (shared bus and cycle lane), code as not present.</p>  <p>[CV first, then if PATH side pass us faded markings, then overwrite]</p> |

| Attribute                                      | Code | Coding guidance                                                                                                                                                                                                                   | Local guidance                                                                                                                                                                                                                 |
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| <p><b>Light segregation</b></p> <p>Present</p> | 1    | Record if there is light segregation between the facility and the road lanes like bollards, flexible posts, or other types of physical object (e.g. a curb) that prevents discretionary vehicle incursion onto a bicycle facility | <p>All our off road paths should have kerb</p>                                                                                                                                                                                 |
| <p>Not present</p>                             | 2    | Record as 'not present' if they are absent, easily traversable (unlikely to discourage/ designed to allow vehicles to cross) or if there are large gaps vehicles can pass through                                                 | <p>TMCR on road cycling lane, West camp road shared bus-bicycle lane, zebra crossing, signalized pedestrian crossing</p>  <p>[CV/rules]</p> |

| Attribute                                                                                 | Code | Coding guidance                                                                                                                                                                                                                                                     | Local guidance                                                                                                                                                                                                                                           |
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| <b>Facility width per direction</b><br>Very Narrow<br>less than 1m per direction<br>(<2m) | 1    | Record the width of the facility for each direction of travel. For 2-way facilities without a centreline, divide the total width by 2<br>Where a centreline is present, but off-centre (i.e 1 direction has more space than the other) record the narrower of the 2 | Footpath 1.5m or 1.8m width<br>[if facility type = footpath + LIDAR FP width <2.0m]<br>If facility type = off road bicycle path + adj sidewalk (0-1m) = absent + LIDAR CYP width <2.0m]                                                                  |
| Narrow<br>1 to 2 m per direction<br>(2-4m)                                                | 2    |                                                                                                                                                                                                                                                                     | Record for all our 2m CYP+1.5m FP, shared paths, 3.6m shared covered linkway<br>[if facility type = off road bicycle path + adj footpath (0-1m)= present + LIDAR CYP width >=2.0m]<br>[if facility type = Multi use path + LIDAR shared path width <=4m] |
| Wide<br>more than 2m per direction<br>(>4m)                                               | 3    |                                                                                                                                                                                                                                                                     | Record for widened 5m shared signalized and zebra ped crossing, 6m pedestrian cum bicycle crossing<br><br>Display the width of footpath or shared/cycling path (if present)<br><br>[GIS/Rule]                                                            |





| Attribute                                 | Code | Coding guidance                                                                                                           | Local guidance                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Adjacent road lane 0-1m</b><br>Present | 1    | Record if there is a road lane with live traffic 0-1m from either side of the facility with no protective barrier present | <p>Record if edge of cycling/ shared path is right next to road kerb <math>\leq 1\text{m}</math>;</p>  <p>At signalized crossing/development access, this refer to the parallel road</p>  |
| Not present                               | 2    |                                                                                                                           | <p>Park connectors that are away from the road, but along canals, in between parks, etc</p>  <p>[CV+rule]</p>                                                                                                                                                              |

| Attribute                                                          | Code       | Coding guidance                                                                                                                                                                                                                                                                                                          | Local guidance                                                                                                                                                                                                                                                                |
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| <b>Adjacent vehicle parking 0-1m</b><br>Present<br><br>Not present | 1<br><br>2 | Record if there is vehicle parking 0-1m from either side of the facility with no protective barrier present                                                                                                                                                                                                              | Datamall, HDB (point) and URA (polygon) car park lots → now using URA car park lots<br><br>GDM/ lanemarking, TYP_CD = E (White-dash-centre line on 2way carriageway) [include double yellow line (no parking) to include illegal parking in private landed area]<br><br>[GIS] |
| <b>Adjacent severe hazard 0-1m</b><br>Present<br><br>Not present   | 1<br><br>2 | Record if there is a severe hazard 0-1m from either side of the facility. Severe hazards include anything which could cause serious injury or death to a bicyclist or light vehicle user and where there is no protective barrier present, such as waterways or deep, open drains or cliffs or significant drop of >60cm | <br><br>[Rare cases/Human intervention]                                                                                                                                                   |

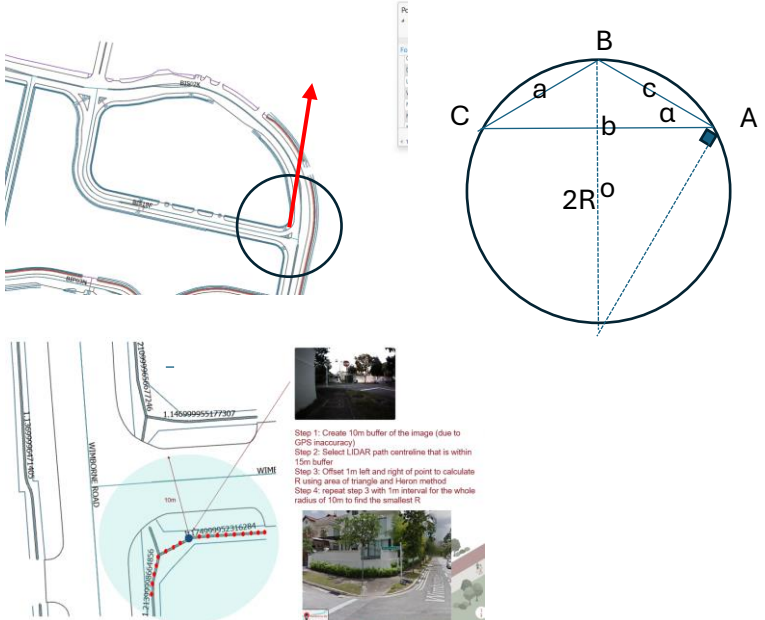
| Attribute                                              | Code | Coding guidance                                                                                                                                          | Local guidance                                                                                                                                                                                                                                                       |
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| <b>Adjacent object or level change 0-1m</b><br>Present | 1    | Record where there is an object or level change $\leq$ 60cm high (such as curb) 0-1m from either side of the facility with no protective barrier present |  <p>If path is right beside road carriageway without railing</p>                                                                                                                  |
| Not present                                            | 2    |                                                                                                                                                          | [CV/Rule]                                                                                                                                                                                                                                                            |
| <b>Adjacent sidewalk 0-1m</b><br>Present               | 1    | Record if there is a pedestrian facility 0-1m from either side of the facility with no physical barrier present (e.g. fence, verge)                      | Record 1 for footpath side by side dedicated cycling path, pedestrian cum bicycle crossing   |
| Not present                                            | 2    |                                                                                                                                                          | Record 2 for signalized widened PC, shared path<br>Record 2 for split path with $>$ 1m of green verge in between<br><br>[CV/Rule]                                                                                                                                    |

| Attribute                                                                                              | Code                                           | Coding guidance                                                                                                                                                                                                                                                           | Local guidance                                                                                                                |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Adjacent road lane 1-3m</b><br>Present<br><br><br><br><br><br><br><br><br><br><br>Not present       | 1<br><br><br><br><br><br><br><br><br><br><br>2 | Record if there is a road lane with live traffic 1-3m from either side of the facility with no protective barrier present.<br><br>Adjacent highways or rural roads with speeds of 80km/h or more (without protective barriers) within 10m of the facility may be recorded | <p>If green verge is 5m, record as 2</p>  |
| <b>Adjacent vehicle parking 1-3m</b><br>Present<br><br><br><br><br><br><br><br><br><br><br>Not present | 1<br><br><br><br><br><br><br><br><br><br><br>2 | Record if there is vehicle parking 1-3m from either side of the facility with no protective barrier present                                                                                                                                                               |                                                                                                                               |

| Attribute                                              | Code | Coding guidance                                                                                                                                                                                                                                                                                                      | Local guidance                                                                        |
|--------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Adjacent severe hazard 1-3m</b><br>Present          | 1    | Record if there is a severe hazard 1-3m from either side of the facility. Severe hazards include anything which could cause serious injury or death to a bicyclist or light vehicle user and where there is no protective barrier present, such as waterways or deep, open drains, or cliffs or significant heights. |    |
| Not present                                            | 2    |                                                                                                                                                                                                                                                                                                                      |                                                                                       |
| <b>Adjacent object or level change 1-3m</b><br>Present | 1    | Record where there is an object or level change $\leq +60$ cm high (such as curb) 1-3m from either side of the facility with no protective barrier present                                                                                                                                                           | Trees                                                                                 |
| Not present                                            | 2    |                                                                                                                                                                                                                                                                                                                      | Railing                                                                               |
| <b>Adjacent sidewalk 1-3m</b><br>Present               | 1    | Record if there is a pedestrian facility 1-3m from either side of the facility with no protective barrier present                                                                                                                                                                                                    | For split paths                                                                       |
| Not present                                            | 2    |                                                                                                                                                                                                                                                                                                                      |  |

| Attribute                  | Code | Coding guidance                                                                           | Local guidance                                              |
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| <b>Grade</b><br><5 degrees | 1    | Record if the facility is on a longitudinal slope of more or less than/equal to 5 degrees | 1:25 (4%, 2 degrees)                                        |
| >=5 degrees                | 2    |                                                                                           | 1:20 (5%, 2.9 degrees)<br><br>[Pending z values from LIDAR] |

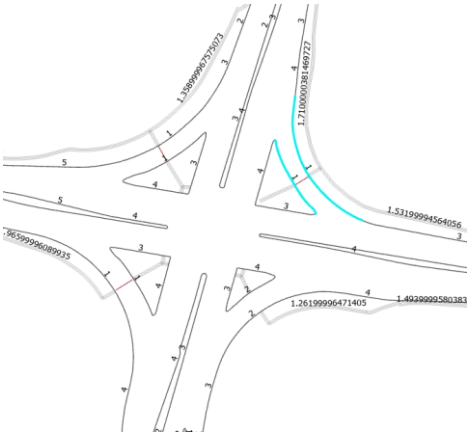


| Attribute                              | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                              | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Curvature</b><br>Sharp turn present | 1    | Record if the facility has a sharp turn present which would require a bicyclist or light vehicle to slow down significantly to negotiate the corner safely <10m or tapering angle >45 degrees. The 10m radius is from the CROW guide, which states that anything below requires significant slowing down to navigate safely (and that below 5m, it is hard to stay upright). |  <p>The local guidance section contains three main visual elements: 1. A map of a road intersection with a red arrow pointing to a sharp turn. 2. A geometric diagram of a circular turn with points A, B, and C on the circumference, and a radius labeled 2R°. 3. A GIS analysis diagram showing a 10m buffer around a road segment, with a list of steps: Step 1: Create 10m buffer of the image (due to GPS inaccuracy); Step 2: Select LQAR path centreline that is within 15m buffer; Step 3: Offset 1m left and right of point to calculate R using area of triangle and Heron method; Step 4: repeat step 3 with 1m interval for the whole radius of 10m to find the smallest R.</p> <p>Area of triangle = <math>\frac{1}{2} bc \sin \alpha</math><br/>           Heron method, Area of triangle = <math>\sqrt{s(s-a)(s-b)(s-c)}</math>,<br/>           where <math>s = \frac{1}{2}(a+b+c)</math><br/> <math>\frac{a}{\sin \alpha} = 2R \sin 90^\circ = 2R</math><br/> <math>R = \frac{a}{2 \sin \alpha}</math><br/>           Need to manually sensor out the U-turning movement of the data collector and also cross paths.<br/>           Buffer reduced further to 5m (to minimize inclusion of opposite paths)</p> <p>[GIS/rules]</p> |
| No sharp turn present                  | 2    |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Attribute                                                                    | Code               | Coding guidance                                                                                                                                        | Local guidance                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Street lighting</b><br>Present<br><br>Not present                         | 1<br><br>2         | Record if the facility has adequate lighting at night                                                                                                  | Default yes                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pedestrian crossing</b><br>Present<br><br><br><br><br><br><br>Not present | 1<br><br><br><br>2 | Record if there is a designated pedestrian crossing or other feature (such as a bus stop) which necessitates pedestrians crossing the bicycle facility | Record for pedestrian gates or possible pedestrian access that interacts with shared path [manual]<br><br>For footpath/CPN/shared path, where there is a bus stop [GIS] and perpendicular walkway connecting to the bus stop<br><br>Also for MRT station exit to bus stops, taxi stands/PUDO/crossings [GIS]<br><br>Road crossing - Signalised Junction/midblock PC/zebra crossing<br><br>[GIS/Manual] |
| <b>Intersecting bicycle facility</b><br>Present<br><br><br>Not present       | 1<br><br>2         | Record if there is an intersecting bicycle facility resulting in cross-flow or more bicycle or light vehicle flow joining the facility                 | Record for intersecting cycling facility<br><br>[Manual]                                                                                                                                                                                                                                                                                                                                               |

| Attribute                                       | Code | Coding guidance                                                                                                                                                                                                                                                             | Local guidance                                                                          |
|-------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Intersection approach</b><br>Shared          | 1    | Record if the approach to a road or street intersection is on-street (shared) 1 or remains separate 2 from traffic until the intersection itself. Eg of shared approaches include facilities ending before the intersection, or where there are pocket lanes for bicyclists |                                                                                         |
| Separate/NA                                     | 2    | Record as NA if no road intersection is present                                                                                                                                                                                                                             | Record 2 for the off road cycling path nearing intersection<br><br>Default: separate/NA |
| <b>Intersection or road crossing</b><br>Present | 1    | Record if there is a road or street intersection or if the facility crosses a road or street                                                                                                                                                                                | Zebra crossing, signalized PC, development accesses, unsignalized junctions             |
| Not present                                     | 2    |                                                                                                                                                                                                                                                                             | [CV]                                                                                    |

| Attribute                              | Code | Coding guidance                                                                                                                                                                                               | Local guidance                                                                                                                                   |
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| <b>Crossing facility</b><br>Present/NA | 1    | Record if there is a safe crossing facility for bicyclists or light vehicles at the intersection or road crossing. Facilities may include signals, pedestrian crossings, raised crossings, refuge islands etc | Zebra crossing, bicycle crossing, Silver zone kerb cut ramp, two-staged crossing, eyeland considered present as there is traffic calming measure |
| Not present                            | 2    |                                                                                                                                                                                                               | Record 2 at development accesses unless development puts in formal crossing, refuge island or kerb cut ramps with traffic calming<br><br>[CV]    |

| Attribute                                                                                 | Code       | Coding guidance                                                                                    | Local guidance                                                                                                                                                                                                                              |
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| <b>Number of lanes – adjacent road</b><br>1 per direction/NA<br><br>>1 per direction      | 1<br><br>2 | Record how many lanes are on the road adjacent to the facility. If there is no road, record as NA  | LIDAR layer: Kerblines<br>Match the nearest Kerblines (lanes) with the photo, Number of lanes – adjacent road (1 or 2)<br><br><br><br>[GIS]              |
| <b>Number of lanes – intersecting road</b><br>1 per direction/ NA<br><br>>1 per direction | 1<br><br>2 | Record how many lanes are on the intersecting road. If there is no road intersection, record as NA | LIDAR layer: Kerblines<br>If Facility Type = Mixed traffic road lane, then match the nearest Kerblines (lanes) with the photo, Number of lanes – Intersecting (1 or 2)<br><br>Else, Number of lanes – intersecting road (1)/NA<br><br>[GIS] |

| Attribute                     | Code | Coding guidance                                                                       | Local guidance   |
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| <b><i>Property access</i></b> |      |                                                                                       |                  |
| <i>Present</i>                | 1    | Record if there are property entrances or driveways which intersect with the facility |                  |
| <i>Not present</i>            | 2    |                                                                                       | [pending CV/GIS] |



| Attribute                                            | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                      | Local guidance                                                                                                                                                                                                                                                   |
|------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Peak pedestrian flow along or across facility</b> |      | <i>Flows and speed data are critical to the model. We could simulate scenarios and identify volume thresholds for risk, but getting data from official sources or collecting samples is always recommended. Pedestrians and bicyclists need to be collected in peak hours on a typical movement day, while vehicle volumes are on a daily basis.</i> |                                                                                                                                                                                                                                                                  |
| None                                                 | 1    | If recorded as 'None', no Bicycle-pedestrian crash type score will be produced)                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                  |
| Low                                                  | 2    | Free flow speed can be maintained during peak periods.                                                                                                                                                                                                                                                                                               | <p>Zebra crossing/signalised PC [auto]<br/>           Perpendicular pedestrian crossing, (CV/rules), bus stops (GDM layer), can also use LOOK marking as a gauge as low</p>  |
| Moderate to high                                     | 3    | Free bicyclist flow is disrupted during peak periods.                                                                                                                                                                                                                                                                                                | <p>High – along schools, hawker centres/town centres/MRT (GDM MRT exit) - 2025/bus interchanges<br/>           For MRT exit points, need to offset 20-25m front and back to include nearby stretches</p> <p>[GIS]</p>                                            |

| Attribute                                                                                         | Code       | Coding guidance                                                                                                                                                                                                                                                             | Local guidance                                                                                                                                  |
|---------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Peak bicycle/ LV traffic flow</b><br><i>Low</i><br><br><i>Moderate to high</i>                 | 1<br><br>2 | Record bicycle/light vehicle flows during peak times, when most activity is present (e.g. morning or evening peak hours or school finish times)<br><br>Steady or busy flows during peak times<br><br>Thresholds can align with >50 bikes/hr or >100 users/day as indicative | AMG's 24/7 sensors (10) with rotation + before and after counts<br><br>[GIS]<br><br>Cyclist, (LV) PMD, PAB;<br>PMA is considered as pedestrians |
| <b>Observed proportion of cargo bikes and mopeds</b><br><i>Low</i><br><br><i>Moderate to high</i> | 1<br><br>2 | Record the proportion of Type B (cargo bike) or D (25-35kph + >35kg) vehicles using the facility                                                                                                                                                                            | D is equivalent to motorcycles as all our ePMD, PAB < 20kg<br><br>Default: low                                                                  |

| Attribute                                                                             | Code                                       | Coding guidance                                                                                                          | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bicycle / LV speed – average</b><br><20km/h<br><br>>= 20km/h                       | 1<br><br><br><br><br><br><br><br><br><br>2 | Record the average free flow speed of bicycles and light vehicles                                                        | Record 1 for normal shared/CPN<br><br><br><br><br><br><br><br><br><br>Record 2 for long, uninterrupted PCN, cycling bridge/underpass (with wide path)<br><br><br>Default 1<br><br><br><br>Future<br>A webpage app developed by AI office, has added object speed detection function (besides the traffic count function), can run speed detection from soe of the sensor sample videos if needed; currently the 'layer' is the tableau dashboard, haven't convert into a shape file or GIS layer yet |
| <b>Bicycle / LV speed differential</b><br><10km/h<br><br><br><br><br><br><br>>=10km/h | 1<br><br><br><br><br><br><br>2             | Record where the standard deviation of the free flow speed of bicycles and light vehicles is greater or less than 10km/h | Default 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Attribute               | Code       | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Local guidance |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|------|------------|---|-------|-------|-------|-----|-------|-------|-------|-----|-------|-------|-------|------|-------|-------|-------|------|-------|-------|-------|------|-------|-------|-------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Road AADT Number</b> | NA         | <p>Record the average annual daily traffic (AADT) of the road (if present). If a road is not present, record it as zero</p> <p><b>Figure 5 CycleRAP AADT and speed risk multipliers</b></p> <table> <tr> <th>AADT</th><th>Multiplier</th><th>AADT</th><th>Multiplier</th></tr> <tr> <td>0</td><td>0.250</td><td>15000</td><td>1.403</td></tr> <tr> <td>100</td><td>0.250</td><td>17500</td><td>1.501</td></tr> <tr> <td>500</td><td>0.500</td><td>20000</td><td>1.606</td></tr> <tr> <td>1500</td><td>0.750</td><td>22500</td><td>1.718</td></tr> <tr> <td>2500</td><td>1.000</td><td>25000</td><td>1.838</td></tr> <tr> <td>5000</td><td>1.070</td><td>30000</td><td>1.967</td></tr> <tr> <td>7500</td><td>1.145</td><td>35000</td><td>2.105</td></tr> <tr> <td>10000</td><td>1.225</td><td>40000</td><td>2.252</td></tr> <tr> <td>12500</td><td>1.311</td><td></td><td></td></tr> </table> | AADT           | Multiplier | AADT | Multiplier | 0 | 0.250 | 15000 | 1.403 | 100 | 0.250 | 17500 | 1.501 | 500 | 0.500 | 20000 | 1.606 | 1500 | 0.750 | 22500 | 1.718 | 2500 | 1.000 | 25000 | 1.838 | 5000 | 1.070 | 30000 | 1.967 | 7500 | 1.145 | 35000 | 2.105 | 10000 | 1.225 | 40000 | 2.252 | 12500 | 1.311 |  |  | <p>To get from iTransport by stretches and reference Mytransport/Traffic news to infer</p> <p>Gotten from Datamall (traffic flow – hourly average traffic flow from a representation month of every quarter during 0700-0900h) → use Postman to access<br/>[However, the data is not islandwide, it is based on available detectors]</p> <p>To get from ERP2 data (7am-9am, any typical weekday)</p> <p>Peak hour *8*350/365= AADT<br/>(PHF is from planning division)</p> <p>2500 (Cat 5 private landed roads)<br/>5000<br/>9000<br/>15000<br/>[If the path is segregated by an underpass then the expressway, exclude the AADT of the expressway]</p> <p>[pending ERP2 data]</p> |
| AADT                    | Multiplier | AADT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Multiplier     |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0                       | 0.250      | 15000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.403          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 100                     | 0.250      | 17500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.501          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 500                     | 0.500      | 20000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.606          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1500                    | 0.750      | 22500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.718          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2500                    | 1.000      | 25000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.838          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5000                    | 1.070      | 30000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.967          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7500                    | 1.145      | 35000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2.105          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 10000                   | 1.225      | 40000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2.252          |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12500                   | 1.311      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |            |      |            |   |       |       |       |     |       |       |       |     |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |      |       |       |       |       |       |       |       |       |       |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Attribute                                                 | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Local guidance                                                                                         |
|-----------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Heavy vehicle flow</b><br><i>Low/restricted access</i> | 1    | Record the level of heavy vehicle flow present on the road. If a road is not present, record it as 'low/restricted access'<br>Streets where heavy vehicles are restricted all or part of the time, may include local traffic zones or specific restrictions. If restrictions are part time, they should correlate with daylight hours and evenings when bicycles and light vehicles are in highest use; Local, neighbourhood streets where there is very low commercial traffic or no bus routes; Country roads or lanes with no heavy vehicle use |                                                                                                        |
| <i>Moderate to high</i>                                   | 2    | A bus route is present along the road; the road is an industrial or commercial area (without restrictions); the road is a major corridor (e.g arterial road or trunk road) or highway (without restriction)                                                                                                                                                                                                                                                                                                                                        | GDM: lane marking layer – markings for bus lanes: A1, A6, A7, L, Q, Q1, Q2, Y (10 Jun 25)<br><br>[GIS] |
| <b>Road speed limit number</b>                            | NA   | Record the road speed limit. If no road is present, record as 'NA'<br>(Speed limit does not affect risk scores. It is used for analysis purpose only.)                                                                                                                                                                                                                                                                                                                                                                                             | From GDM layer [2015]<br><br>[GIS]                                                                     |

| Attribute                                 | Code | Coding guidance                                                                                                                                                                                                                                                                                                                                                                                                      | Local guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Road operating speed (mean) number</b> | NA   | <p>Record the mean operating speed of motorized traffic on the road. If no road is present, record as 'NA'</p> <p>Average speed at which motorized vehicles travel in <b>free flow</b> conditions along the road. Mean speed may be smoothed along homogenous lengths of road, but should reflect the changes in operating speeds due to traffic calming measures, intersection design and other design features</p> | <p>To get average speed for some major arterial roads from iTransport.</p> <p>Then use an off peak MyTransport app/Traffic news red-amber-green heatmap as a proxy to scale the speeds for other roads</p> <p>From Datamall: Traffic Speed Bands → use Postman to access<br/>[Data is not complete, not islandwide]</p> <p>From ERP2 data → Any weekday 10-11am [Need to cater for OBU adoption rate]</p> <p>For 50kph roads, midblock: 45kph, at development access &amp; 1 frame after sign. intersection: 40kph, slightly before and after hump, signalized intersection: 30kph<br/>If speed &lt;1k/h, there appears to be some errors.</p> <p>[GIS]</p> |

## Grouping appearance of attributes

|                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Facility configuration</u></b><br>Area type<br>Facility type<br>Adjacent sidewalk 0-1m<br>Adjacent sidewalk 1-3m<br>Adjacent road lane 0-1m<br>Adjacent road lane 1-3m<br>Adjacent vehicle parking 0-1m<br>Adjacent vehicle parking 1-3m<br>Adjacent object or level change 0-1m<br>Adjacent object or level change 1-3m | <b><u>Facility clear width</u></b><br>Facility Access<br>Fixed obstacle on facility<br>Non-fixed obstacle on facility<br>Facility width per direction<br>Width restrictions<br>Light segregation<br>Adjacent severe hazard 0-1m<br>Adjacent severe hazard 1-3m<br>Line of sight | <b><u>Facility surface conditions</u></b><br>Delineation<br>Major surface road deformation<br>Loose or slippery surface<br>Grade<br>Curvature<br>Tram or train rails<br>Street lighting |
| <b><u>Intersection</u></b><br>Intersection approach<br>Intersection or road crossing<br>Crossing facility<br>Property access<br>Pedestrian crossing<br>Intersecting bicycle facility<br>Number of lanes – adjacent road<br>Number of lanes – intersecting road                                                                 | <b><u>Flow &amp; Speed</u></b><br>Flow direction<br>Peak pedestrian flow along or across<br>Peak bicycle/LV traffic flow<br>Obs proportion of cargo bikes<br>Bicycle/LV speed average<br>Bicycle/LV speed differential<br>Road AADT<br>Road Operating speed (mean)              |                                                                                                                                                                                         |



## GIS Shp file layers used in tool

- MRT\_EXITS.shp - Peak pedestrian flow along, across facility, pedestrian crossing
- Bus lanes.shp - Heavy vehicle flow
- URA\_PARKING\_LOT.shp - Adjacent Vehicle Parking 0-1m
- LanduseRural2025.shp (rural) - width and curvature
- LanduseIndustrial2025.shp (industrial) - width and curvature
- CentralMB2025.shp (inner\_urban) - width and curvature
- ROADATTRIBUTELINE\_SPEEDLIMITS.shp - road speed limit
- 31Oct24\_Link\_FUL.shp - Road Operating Speed
- Bus\_Stop.shp – pedestrian crossing
- Road\_crossing.shp – Pedestrian crossing

1) After computing the sheet, it would be good to relook at the coding again to adjust for relative comparison across segments or to clean up incorrect entry in the Upload\_data sheet directly

2) Calculate the CycleRAP score

VB crashes:

- Low risk: <10 (displayed in green)
- Medium risk: 10 to 25 (displayed in yellow)
- High risk: 25 to 60 (displayed in red), and
- Extreme risk: > 60 (displayed in dark red).

Other types of crashes (BB, BP, SB)

- Low risk: <5 (displayed in green)
- Medium risk: 5 to 10 (displayed in yellow)
- High risk: 10 to 20 (displayed in red), and
- Extreme risk: >20 (displayed in dark red).

The CycleRAP score (i.e. the sum of the crash types) is displayed as being low, medium, high or extreme risk only where one or more of the crash types fall into that category. That is, the CycleRAP score will only be classified as 'Medium risk' if one or more of the crash types present medium risk scores, and 'High risk' if one or more of the crash types present high risk scores.